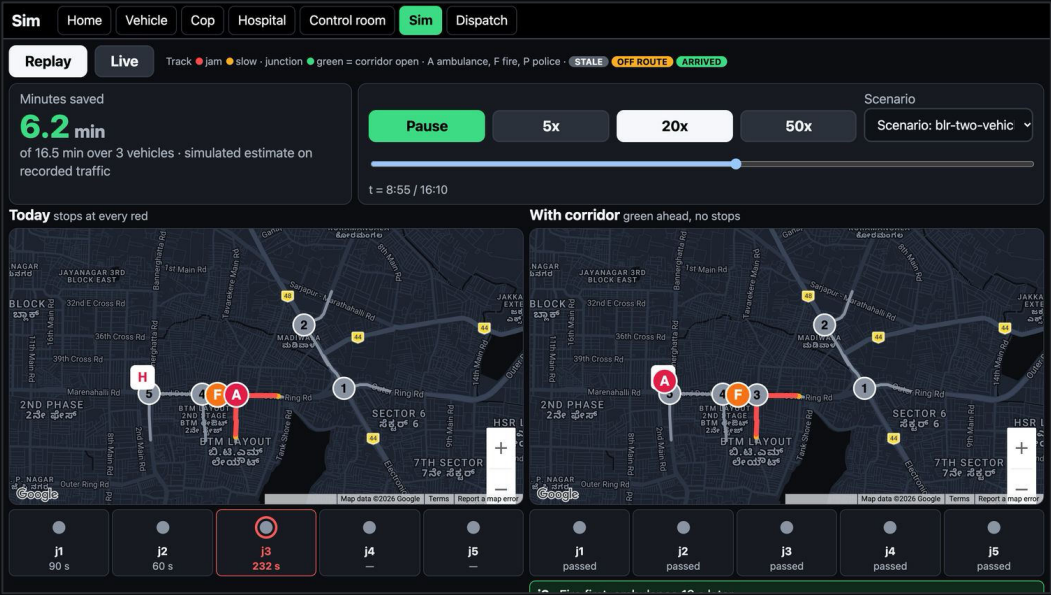


Emergency Green Corridor

Warning the junction constable before the siren arrives.

Team Green Corridor · Nandish · green-corridor-2026.web.app



Team details

A solo entry, built end to end by one contributor.

TEAM NAME

Green Corridor

TEAM LEADER

Nandish

CONTRIBUTORS

Sole contributor

PROBLEM STATEMENT

Sustainability & Social Impact: emergency vehicles stuck at red signals while the junction constable is never told.

Brief about the idea

Warn the junction constable early, sized to the queue

16.5 min

saved on the scripted scenario

4 min

warning at a 400 m queue

100 %

tier accuracy on 11 synthetic clips

Scripted scenario and synthetic voices, not field results.

Opportunities (how it differs / how it solves / USP)

Not a fixed radius, and not waiting for new hardware



HOW IT DIFFERS

**Warning sized to
the queue, not a
fixed radius**



HOW IT SOLVES

**The constable
clears the queue
before the siren**



USP

**Rules decide,
Gemini explains;
no hardware
needed**

List of features

Six screens and the engines behind them

Suggested tier

CRITICAL

Crew confirms

Confirm CRITICAL

Voice triage

STOP

ESCALATED

Stop cross traffic immediately.

Spoken cop alert

Vitals

SBP

85

SBP 86 → 85

HR

112

SpO2

90

SpO2 89 → 90

Treatment so far: Aspirin 300 mg

Hospital live brief

Control room board

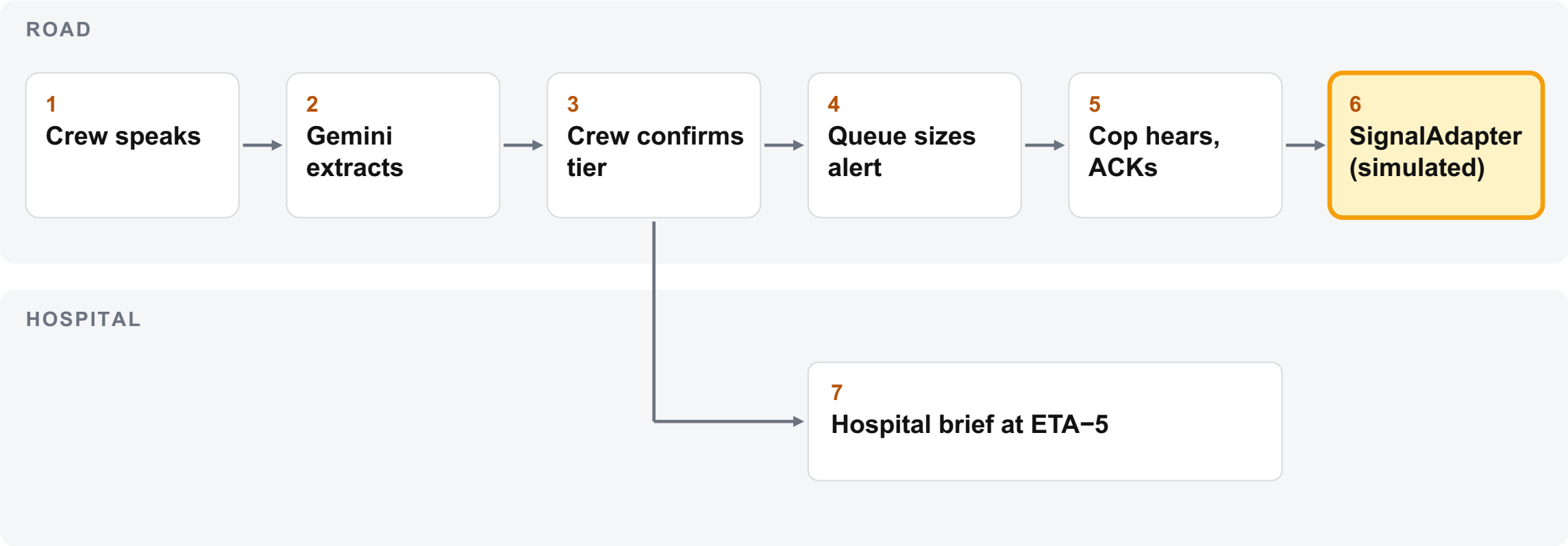
With-vs-without replay

Routing agent trace

- Lead time
- Sequencing
- Cop voice
- ATMIST
- After-action
- Eval harness

Process flow

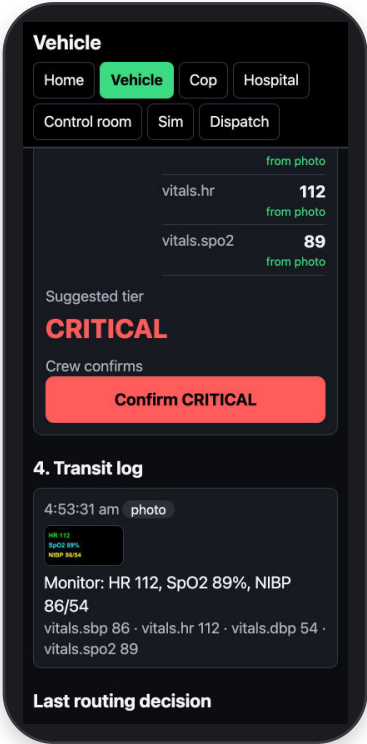
From the crew’s words to the constable’s phone



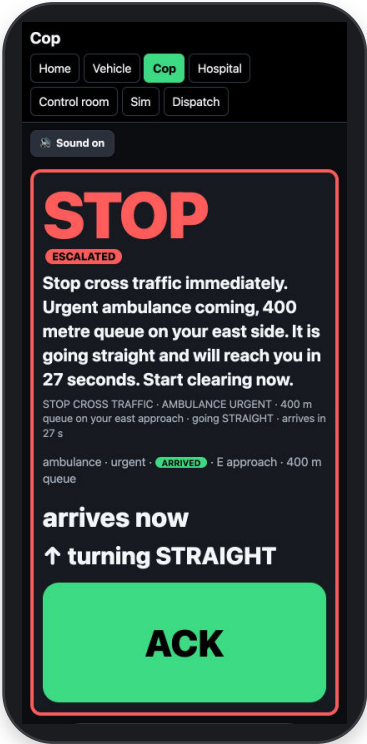
Amber seam: simulated today.

Wireframes

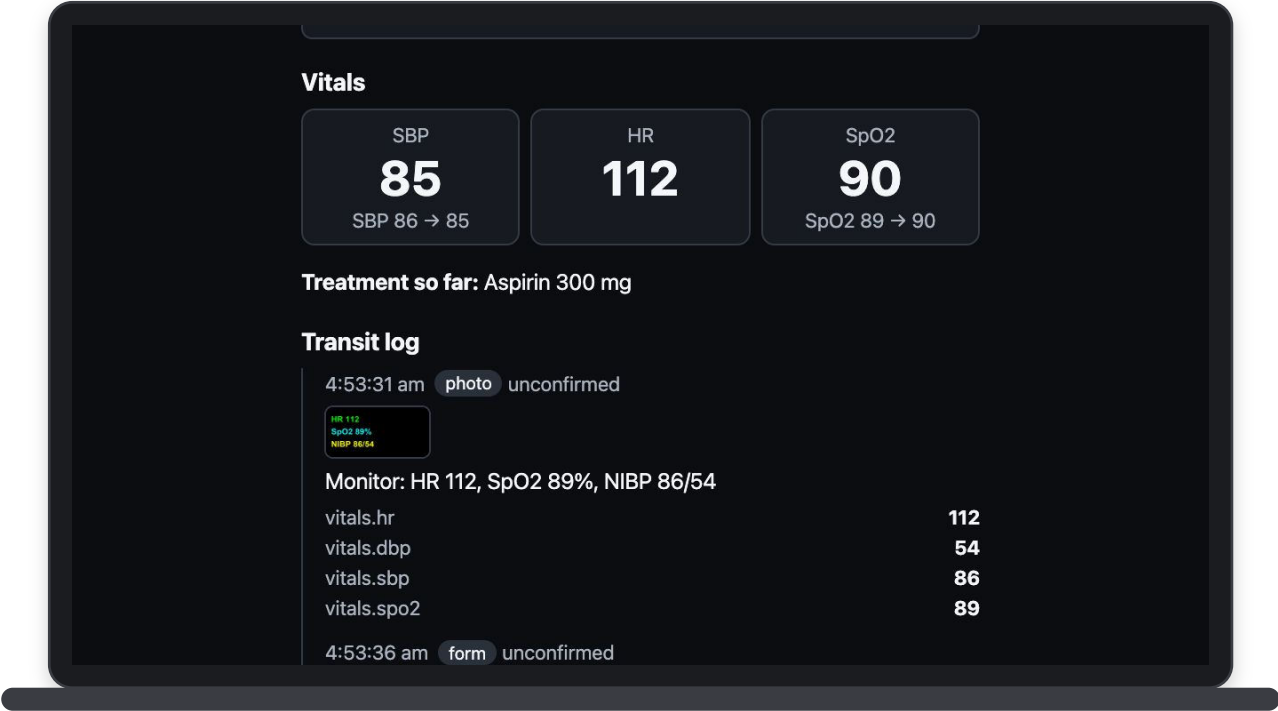
Real screens from the running prototype



Voice triage



Spoken alert

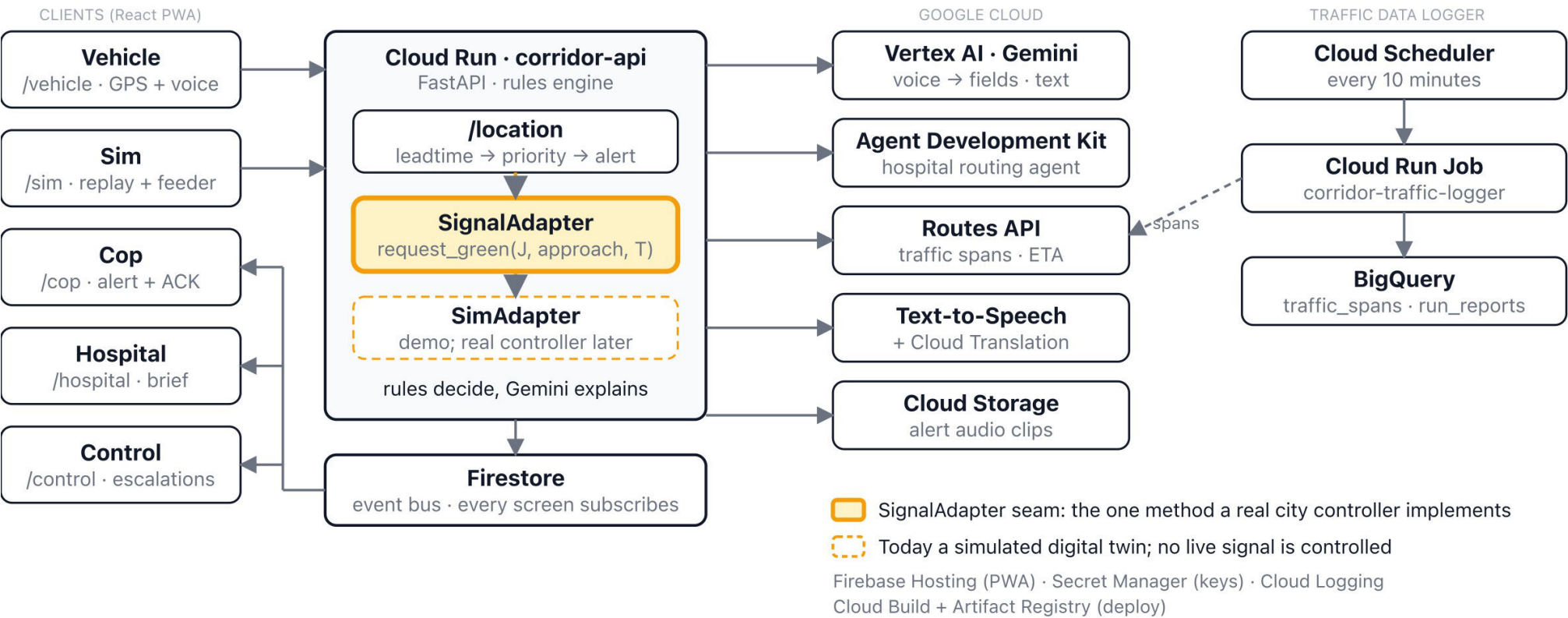


Hospital brief

Live rehearsal captures; synthetic patients.

Architecture

Rules decide; Google Cloud services extract and explain



Rules engine on Cloud Run, Firestore event bus; the amber SignalAdapter seam is where a real controller plugs in.

Technologies (Google Cloud)

A Google Cloud stack, from model to delivery

● AI	Gemini on Vertex AI	Google ADK	BigQuery ML
● Run & data	Cloud Run	Firestore	BigQuery
● Maps & speech	Routes API	Text-to-Speech	Translation
● Delivery	Firebase Hosting	Cloud Build	Secret Manager

Also: Storage, Scheduler, Logging

Estimated implementation cost

About ₹155 per vehicle-hour, almost all of it Routes

Service	Basis	₹ per hour
Routes	3 calls/min × 60 × ₹0.85	₹153
Gemini	17 calls × ₹0.02	₹0.34
Text-to-Speech	~1,000 chars × ₹1.30/1,000	₹1.30

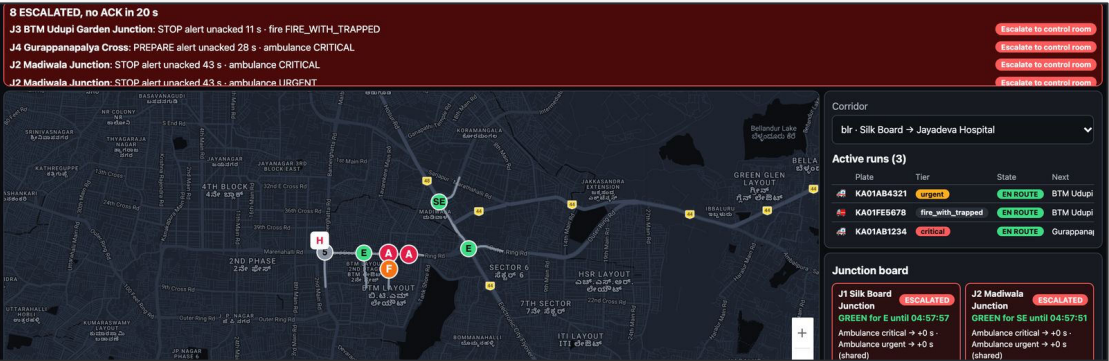
≈ ₹155

per vehicle-hour (estimate)

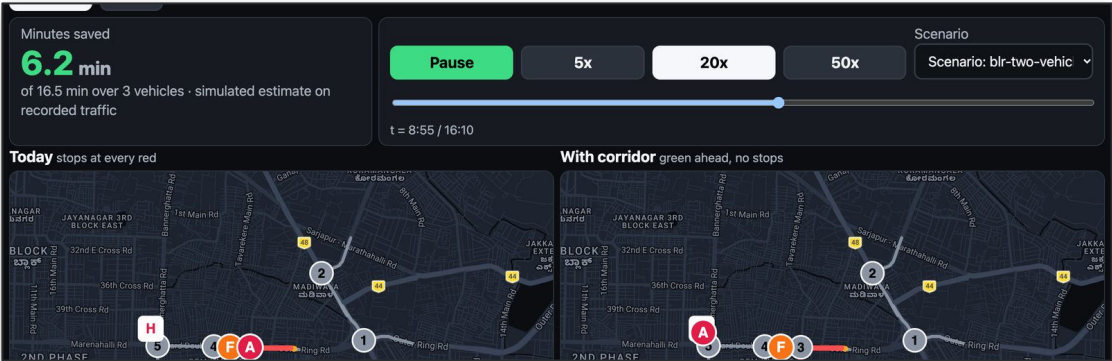
List prices; Routes is ~98 %.

Snapshots of the prototype

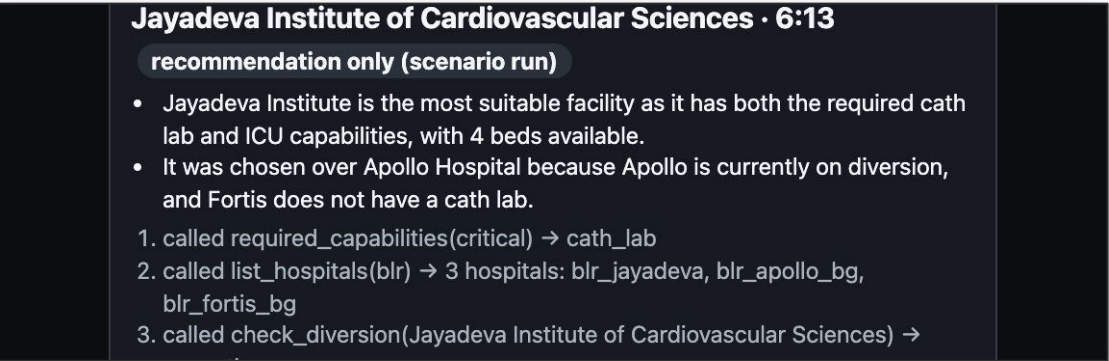
The loop, mid-run, from the live rehearsal



Control room



Replay simulator



Routing trace



Hospital log

Snapshots of the prototype (routing agent)

Live routing trace: Jayadeva chosen, two alternatives rejected

```
1 required_capabilities(critical) -> cath_lab
2 list_hospitals(blr) -> 3 hospitals
3 check_diversion(Jayadeva) -> accepting
4 check_diversion(Apollo) -> on diversion
5 eta_to(Jayadeva) -> 373 s, 2.6 km
6 Added to baseline: icu
```

Destination Jayadeva, confidence 1.0

Jayadeva Institute of Cardiovascular Sciences · 6:13

recommendation only (scenario run)

- Jayadeva Institute is the most suitable facility as it has both the required cath lab and ICU capabilities, with 4 beds available.
- It was chosen over Apollo Hospital because Apollo is currently on diversion, and Fortis does not have a cath lab.

```
1. called required_capabilities(critical) -> cath_lab
2. called list_hospitals(blr) -> 3 hospitals: blr_jayadeva, blr_apollo_bg,
   blr_fortis_bg
3. called check_diversion(Jayadeva Institute of Cardiovascular Sciences) ->
   accepting
4. called check_diversion(Apollo Hospital Bannerghatta Road) -> on diversion
5. called eta_to(Jayadeva Institute of Cardiovascular Sciences) -> 373 s, 2.6 km
6. capabilities: cath_lab (Required for immediate intervention for suspected
   STEMI/ACS.); icu (Required for patient with cardiogenic shock (hypotension,
   tachycardia) and respiratory distress.). Added to baseline: icu
```

The trace as shown in /vehicle.

Performance report / benchmarking

Synthetic-voice results, scripted-run timings

90 %

field accuracy

100 %

tier accuracy

3.5_s

mean extraction

4 vs 1

min lead, long vs short

11 synthetic clips, not field data. Before the fix: 87 % / 73 %.

17 / 37_{ms}

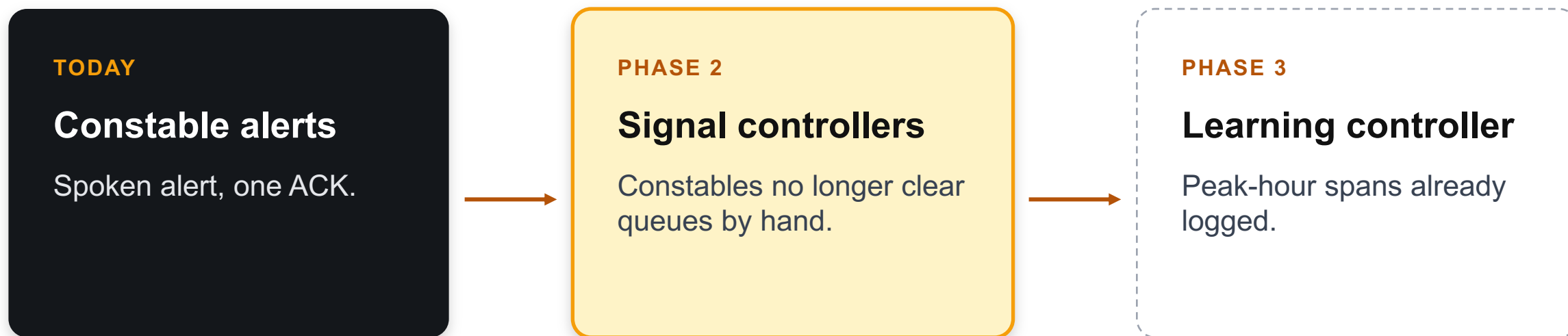
/location p50 / p95, offline

344 Python · **138** web tests

98 % coverage

Additional details / future development

From constable to controller, in three steps



Signals simulated, patients synthetic, minutes saved estimated.

Links

Try it, read the code, watch the demo

GITHUB PUBLIC REPOSITORY



github.com/Nandish3010/ideal-disco

DEMO VIDEO (3 MINUTES)

[link]

FINAL PRODUCT



green-corridor-2026.web.app

Warn the constable before the siren arrives.

Try it: green-corridor-2026.web.app

Read it: github.com/Nandish3010/ideal-disco

Built by Nandish